ALICE L WALTON SCHOOL OF MEDICINE

Spectrum of Activity and Indications–3 Fluoroquinolones and Sulfonamides

Last updated June 13, 2025 5 min read CC BY-SA 4.0

Authors: Diptiman Bose, Trager Hintze, Ray Brown
Editors: Jordan Finn, Russell Johnson

Listen to this Brick
0:00 / 10:37 1x

Learning Objectives (1)

After completing this brick, you will be able to:

1. Describe the spectrum of activity of fluoroquinolones and sulfonamides.

CASE CONNECTION

KO, a 22-year-old female, presents with symptoms of acute cystitis that have persisted for 2 days. She reports experiencing dysuria, increased urinary frequency, and urgency. Additionally, she feels a sensation of incomplete bladder emptying and mild lower abdominal discomfort. KO denies having a fever. She states that she is not pregnant. She states that she has a known allergy to fluoroquinolones, which causes major hives. She says that during her last UTI occurrence, 8 months back, she has a bad skin reaction to sulfamethoxazole/trimethoprim.

Upon physical examination, her vital signs are normal, and there is mild suprapubic tenderness without costovertebral angle tenderness. There are no signs of systemic infection. You are a medical student on rotation. You confirm with your preceptor that KO has the signs and symptoms of acute cystitis, likely caused by a urinary tract infection, typically caused by Gram Negative Rods of the Enterobacteriaceae family.

Your preceptor asks you the possible treatment options for KO? Consider your answer as you read, and we'll revisit KO at the end of the brick.

GO TO CONCLUSION ↓

We are going to discuss the spectrum of coverage of Fluoroquinolones and sulfonamides. First, let's recall some key details about these agents.

ACTIVE LEARNING QUESTION

Which of the following adverse effects/warnings are associated with the use of fluoroquinolones (FQ)?

A. ☐ Red-man syndrome
B. ☑ QTc prolongation
C. ☑ Tendon rupture
D. ☑ Avoid in pregnancy
E. ☑ Photosensitivity

Explanation

The use of FQ is associated with QTc prolongation (B) and the risk of increasing QT interval should be avoided. The risk of tendon rupture (C) increases with age >60. FQ are teratogenic and should be avoided in pregnancy (D). FQ can cause photosensitivity (E).Red man syndrome (A) is associated with the use of Vancomycin and not the fluoroquinolones.

What pathogens can the fluoroquinolones cover? (LO#1)

Fluoroquinolones are broad-spectrum antibiotics. They have activity against gram-positive, gram-[illegible] However, there are very specific distinctions between the spectrum of activity between the agents.

Ciprofloxacin: Ciprofloxacin has the greatest activity against aerobic gram-negative bacilli. These include *Escherichia coli, Klebsiella* spp, *Proteus* spp. This makes it a good option for intraabdominal infections and urinary tract infections. It also has potent activity for *H. influenzae, M. catarrhalis.* It also has activity against *Pseudomonas aeruginosa*. Ciprofloxacin has limited coverage for gram-positive bacteria (*S. pneumoniae, Staphylococcus aureus).*

Levofloxacin: Levofloxacin has activity against both gram-positive and gram-negative pathogens. It also has activity against *P. aeruginosa*. Levofloxacin is also active against most common respiratory pathogens, including *Streptococcus pneumoniae, Haemophilus influenzae*, and *Moraxella catarrhalis*.

Which fluoroquinolone has the greatest activity against aerobic gram-negative bacilli?

Moxifloxacin: Moxifl[illegible]imilar to Levofloxacin. It also has coverage against the respiratory pathogens *Streptococcus pneumoniae, Haemophilus influenzae*, and *Moraxella catarrhalis*. It also has activity against *P. aeruginosa*. However, Moxifloxacin has enhanced gram-positive activity, making it effective against polymicrobial infections such as intra-abdominal infections.

Which fluoroquinolone has enhanced gram-positive activity?

What pathogens does sulfamethoxazole/trimethoprim (SMX/TMP) cover? (LO#1)

Sulfonamides cover some gram-positive and gram-negative bacteria. Gram-positive coverage includes *Staphylococcus aureus* and *Staphylococcus epidermides*. They are effective against gram-negative bacteria, including *Escherichia coli, Klebsiella pneumoniae*, *Haemophilus influenzae*, and *Proteus mirabilis*. The combination of SMX/TMP is commonly used to treat UTIs. Sulfonamides are also active against certain protozoa Toxoplasma species, such as *Toxoplasmosis gondii*, which can cause sexually transmitted infections (STIs). However, their clinical usefulness has decreased over time due to the emergence of bacterial resistance. Additionally, topical sulfonamides are used to treat burns and bacterial conjunctivitis.

The table below provides a summary of many of the medications and pathogens we have covered so far!

Pathogens	Antibiotic Regimen	Coverage
Non MRSA gram-positive (*S. aureus, Group A Streptococcus, S. epidermidis*)	Ampicillin/Sulbactam, Ertapenem	Good gram-positive coverage
MRSA	Vancomycin, Linezolid	MRSA coverage
Gram-negative *E. Coli, Klebsiella pneumoniae, Proteus mirabilis*	Ertapenem, Levofloxacin/Moxifloxacin	Good gram-positive and negative coverage
	Ceftriaxone	Gram-negative coverage (no pseudomonas)
Pseudomonas	Cefepime, Piperacillin/Tazobactam/Carbapenems (except Ertapenem)	Gram-negative and pseudomonal coverage

Table 1

CLINICAL CORRELATION

Acute cystitis is an [illegible] inary tract. Acute cystitis is primarily caused by the bacterium Escherichia coli *(E. coli)*. Other pathogens that can cause acute cystitis include *Klebsiella, Proteus, Enterococci,* and *Staphylococcus saprophyticus.* The treatment for acute cystitis typically involves antibiotics. First-line treatment of UTI is often **sulfamethoxazole/trimethoprim**. Fluoroquinolones such as ciprofloxacin and levofloxacin can also be used.

CLINICAL REASONING

Why is Moxifloxacin not used in UTI infection? Moxifloxacin is not typically used for treating urinary tract infections (UTIs) because it is not renally eliminated and does not concentrate in the urine. This means that it does not reach effective levels in the urinary tract to combat the bacteria causing the infection.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Based on your reading, what is your recommendation for KO?

Your physical examination and evaluation of KO shows no signs of systemic infection. The assessment indicates acute cystitis, likely caused by a urinary tract infection. Urine cultures would confirm the pathogen causing the infection. Based on her last culture results, you can initiate an empiric therapy.
Sulfamethoxazole/Trimethoprim could be considered. However, KO has a sulfa allergy. The use of β-lactams such as amoxicillin/Clavulanate or cephalexin would be good options to consider. KO should complete the full course of antibiotics and avoid irritants such a[illegible]vent exacerbation of symptoms and future UTIs.

Summary

- Fluoroquinolones are broad-spectrum antibiotics effective against gram-positive, gram-negative, and atypical pathogens, with specific activity distinctions among agents.
- Ciprofloxacin is most effective against aerobic gram-negative bacilli, while Moxifloxacin has enhanced gram-positive activity.
- Sulfonamides, including sulfamethoxazole/trimethoprim, cover some gram-positive and gram-negative bacteria and are commonly used for UTIs, though resistance is a concern.

Review Questions

1. Ciprofloxacin will effectively cover which of the following:

A. ○ *S. pneumoniae, E. Coli, P. aeruginosa*
B. ○ *S. aureus, K. pneumoniae, P. aeruginosa*
C. ○ *S. pneumoniae, S. aureus, K. pneumoniae*
D. ● *K. pneumoniae, E. Coli, P. aeruginosa*

Hide Explanation

The correct answer is *K. pneumoniae, E. coli, and P. aeruginosa* (D). Fluoroquinolones generally have effective broad-spectrum gram-positive and gram-negative coverage; however, there are differences between agents. Ciprofloxacin, for instance, has good gram-negative coverage including *K. pneumoniae, E. coli, and P. aeruginosa*, but does not effectively cover gram-positive pathogens such as *S. pneumoniae and S. aureus (Choices A, B, and C).*

2. Which antibiotic has enhanced gram-positive activity?

A. ○ Ciprofloxacin
B. ○ Levofloxacin
C. ● Moxifloxacin

Hide Explanation

The correct answer is Moxifloxacin (C). Moxifloxacin has enhanced gram-positive activity, making it effective against polymicrobial infections such as intra-abdominal infections. Ciprofloxacin (A) has activity against both gram-positive and gram-negative pathogens but does not have enhanced gram-positive activity. Levofloxacin (B) also had limited gram-positive coverage.

Hide Explanation

The correct answer is Moxifloxacin (C). Moxifloxacin has enhanced gram-positive activity, making it effective against polymicrobial infections such as intra-abdominal infections. Ciprofloxacin (A) has activity against both gram-positive and gram-negative pathogens but does not have enhanced gram-positive activity. Levofloxacin (B) also had limited gram-positive coverage.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the spectrum of activity of fluoroquinolones and sulfonamides.~~

Objective Confidence: 1/1